

Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow creators to submit draft media deliverables for client review and release locked milestone payments upon sponsor approval sign-off.	High	Pass: Milestone marked completed and funds credited to creator balance. Fail: Funds released without client sign-off.	Core escrow trust mechanism — payment must only follow verified deliverable approval.
FR-002	Functional	The system shall allow the Content Creator and Client Sponsor to jointly define a deliverable milestone (title, due date, locked payment amount) before any draft work is submitted.	High	Pass: A milestone record exists with title, due date, and locked payment amount visible to both parties prior to draft submission. Fail: A draft can be submitted with no associated milestone.	Establishes the contractual basis for payment release and prevents scope/payment disputes later.
FR-003	Functional	The system shall automatically apply a visible watermark to any draft media asset uploaded by the Content Creator before it is viewable by the Client Sponsor.	High	Pass: Every draft opened by the sponsor displays a watermark overlay marking it as unreleased. Fail: An unwatermarked draft is viewable by the sponsor.	Protects the creator's IP from unauthorized use before payment is finalized.
FR-004	Functional	The system shall allow the Client Sponsor to request a revision on a submitted draft, including written feedback, instead of approving the milestone.	Medium	Pass: A revision request with feedback text is logged against the milestone and the Creator is notified within 1 minute. Fail: A milestone can be approved while a revision request is still open.	Supports iterative collaboration between the two parties without releasing funds prematurely.
FR-005	Functional	The system shall allow either the Content Creator or the Client Sponsor to raise a dispute on a milestone, which freezes the associated escrow funds pending admin review.	Medium	Pass: Raising a dispute sets milestone status to "Disputed" and blocks the Release Payment action. Fail: Payment can be released while a milestone is in "Disputed" status.	Provides a safeguard for contested deliverables, consistent with the trust guarantees of an escrow platform.
NFR-001	Nonfunctional	Type: Performance & Security. Uploaded media drafts must process automated watermarking in under 5 seconds for video/image previews.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load. Fail: Watermarking exceeds 5s or raw (unwatermarked) assets are exposed during processing.	Prevents workflow bottlenecks for creators and ensures unreleased assets are never exposed unprotected, even momentarily.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-002	Nonfunctional	Type: Security. The system shall encrypt all escrow payment transaction data, both in transit and at rest, using industry-standard encryption (TLS 1.2+ for transit, AES-256 for storage).	High	Pass: Security audit / penetration test confirms no payment data is transmitted or stored unencrypted. Fail: Any payment field is found in plaintext in transit or in storage.	Protects sensitive financial data — a baseline compliance and trust requirement for a platform that directly custodies milestone payments.

USE CASE SCENARIO

Use Case: Approve Milestone & Release Escrow Payment

Primary Actor: Client Sponsor

Supporting Actor: Content Creator

Purpose: Allow the Client Sponsor to approve a submitted milestone and release the corresponding locked escrow payment to the Content Creator.

Preconditions

A project and milestone have been defined between the Content Creator and Client Sponsor.

The Content Creator has submitted a draft deliverable for the milestone.

The milestone payment is locked in escrow.

The Client Sponsor is authenticated and has access to the project.

Postconditions

If approval succeeds, the milestone is marked as approved/paid and the escrow payment is released to the Content Creator.

The payment-release event and sponsor approval are recorded by the system.

If approval does not occur, the escrow funds remain locked.

Main Success Scenario

1. The Client Sponsor opens the submitted milestone.
2. The system displays the watermarked draft asset and milestone details.
3. The Client Sponsor reviews the draft deliverable.
4. The Client Sponsor selects the approval option.
5. The system records the sponsor's approval sign-off.
6. The system verifies that the approval is associated with the correct milestone and authorized sponsor.
7. The system releases the locked milestone payment to the Content Creator.
8. The system updates the milestone status and records the payment-release event.
9. The Content Creator is notified that the milestone payment has been released.
10. The use case ends successfully.

Alternate Flow — Revision Required

- 4a. The Client Sponsor selects Request Revision instead of approving the milestone.
 - 4a1. The system prompts the Client Sponsor to provide revision feedback.
 - 4a2. The sponsor submits the revision feedback.
 - 4a3. The system records the feedback and marks the milestone as Revision Requested.
 - 4a4. The escrow payment remains locked and the Content Creator is notified.
 - 4a5. The Content Creator submits a revised draft for review.

Relationship Used in the UML Model

Approve Milestone & Release Escrow Payment «include» Process Milestone Payment, because payment processing is required before the locked milestone funds can be released.

UML Diagram

UML Use-Case Diagram: Freelance Content Creator Escrow Platform

PES University Dept. of CSE | Problem Statement #53 (Media, Events & Community)

